CARLETON UNIVERSITY

Department of Mechanical and Aerospace Engineering

Course : MAAE 3500

Project: Motor

April 10 , 2025

Student No : 101203558

David Anyiam

Email: DavidAnyiam@cmail.carleton.ca

Signature : D.A

Question 1

Assuming 100% efficiency, the electrical power input equals the mechanical power output:

$$e_m i_m = \tau_m \omega_m$$

From the project statement,

$$\omega_m = K_\omega e_m \text{and} \tau_m = K_T i_m$$

Rearrange the speed equation:

$$e_m = \frac{\omega_m}{K_\omega}$$

Substitute into the power equation:

$$\left(\frac{\omega_m}{K_\omega}\right) i_m = \tau_m \omega_m$$

Now substitute

$$\tau_m = K_T i_m$$

so:

$$\left(\frac{\omega_m}{K_\omega}\right) i_m = (K_T i_m)\omega_m$$

Cancel $i_m$and ω_m:

$$\frac{1}{K_\omega} = K_T$$

So the theoretical relationship is:

$$\boxed{K_T = \frac{1}{K_\omega}}$$

This is only directly valid when:

- $K_\omega$is in rad/s/V
- $K_T$is in N m/A.

The datasheet gives:

- $K_\omega$in rpm/V
- $K_T$in mNm/A.

So unit conversion is needed.

Unit conversion

$$K_\omega(\text{rad/s/V}) = K_\omega(\text{rpm/V})\left(\frac{2\pi}{60}\right)$$

Then

$$K_T(\text{N m/A}) = \frac{1}{K_\omega(\text{rad/s/V})}$$

and

$$K_T(\text{mNm/A}) = 1000\, K_T(\text{N m/A})$$

Verification using the first motor column

From the first column of the datasheet:

$$K_\omega = 270 \text{ rpm/V}$$

and the listed torque constant is:

$$K_T = 35.3 \text{ mNm/A}$$

Now calculate it from theory.

Convert K_ω to rad/s/V

$$K_\omega = 270\left(\frac{2\pi}{60}\right)$$
$$K_\omega = 270(0.10472) = 28.274 \text{ rad/s*V}$$

Use $K_T = 1/K_\omega$

$$K_T = \frac{1}{28.274} = 0.03537 \text{ N m/A}$$

Convert to mNm/A

$$K_T = 0.03537 \times 1000 = 35.37 \text{ mNm/A}$$

Compare with datasheet

Calculated:

$$K_T = 35.37 \text{ mNm/A}$$

Datasheet:

$$K_T = 35.3 \text{ mNm/A}$$

These agree very closely, so the datasheet values are consistent with the theoretical ideal-motor relationship. The small difference is due to rounding.

Assuming 100% motor efficiency, the electrical input power equals the mechanical output power, so $e_m i_m = \tau_m \omega_m$. Using $\omega_m = K_\omega e_m$ and $\tau_m = K_T i_m$, the motor constants are related by $K_T = 1/K_\omega$, provided SI units are used. For the selected 18 V motor, the datasheet gives $K_\omega = 270$rpm/V and $K_T = 35.3$mNm/A. After converting K_ω to rad/s*V, the theoretical torque constant is found to be 35.37mNm/A, which agrees very closely with the datasheet value. This confirms that the specified motor constants are consistent with theoretical expectations.

Question 2

Select a gear ratio that ensures the mixer operates at 100 RPM when the motor runs at its nominal speed.

$$N_g = \frac{\omega_{out}}{\omega_m}$$

For the selected first motor column, the nominal speed is:

$$\omega_m = 4360 \text{ rpm}$$

The required mixer speed is:

$$\omega_{out} = 100 \text{ rpm}$$

Substitute into the equation:

$$N_g = \frac{100}{4360}$$

$$N_g = 0.02294$$

So the required gear ratio is:

$$\boxed{N_g = 0.02294}$$

This means the output speed is 2.294% of the motor speed. It is equivalent to a speed reduction of:

$$\frac{1}{0.02294} = 43.6$$

Question 3

Using the gear ratio found in Question 2,

$$N_g = \frac{\omega_{out}}{\omega_m} = 0.02294$$

and

$$N_g = \frac{\tau_m}{\tau_{out}}$$

so

$$\tau_{out} = \frac{\tau_m}{N_g}$$

The load torque on the mixer is modeled as viscous damping:

$$\tau_{out} = b\,\omega_{out}$$

Therefore,

$$b = \frac{\tau_{out}}{\omega_{out}}$$

We now do this twice:

- once using the maximum continuous motor torque
- once using the stall torque

For the selected 18 V motor column, the datasheet gives:

- nominal torque (max continuous torque) = 388 mNm
- stall torque = 11500 mNm
- required mixer speed = 100 rpm.

Maximum damping for continuous current

At the continuous-current limit, the motor can provide its maximum continuous torque:

$$\tau_m = 388 \text{ mNm} = 0.388 \text{ N m}$$

Using

$$\tau_{out} = \frac{\tau_m}{N_g}$$

gives

$$\tau_{out} = \frac{0.388}{0.02294} = 16.92 \text{ N m}$$

Now convert mixer speed to rad/s:

$$\omega_{out} = 100\left(\frac{2\pi}{60}\right) = 10.472 \text{ rad/s}$$

Then

$$b_{max} = \frac{\tau_{out}}{\omega_{out}}$$
$$b_{max} = \frac{16.92}{10.472} = 1.615 \text{ N m s/rad}$$

So, the maximum damping that keeps the motor within its maximum continuous current is

$$\boxed{b_{max} = 1.62 \text{ N m s/rad}}$$

Damping corresponding to stall torque

From the datasheet,

$$\tau_{stall} = 11500 \text{ mNm} = 11.5 \text{ N m}$$

Again,

$$\tau_{out,stall} = \frac{\tau_{stall}}{N_g}$$

so

$$\tau_{out,stall} = \frac{11.5}{0.02294} = 501.4 \text{ N m}$$

Now compute the corresponding damping:

$$b_{stall} = \frac{\tau_{out,stall}}{\omega_{out}}$$
$$b_{stall} = \frac{501.4}{10.472} = 47.88 \text{ N m s/rad}$$

Thus,

$$\boxed{b_{stall} = 47.9 \text{ N m s/rad}}$$

Final answers

$$\boxed{b_{max} = 1.62 \text{ N m s/rad}}$$
$$\boxed{b_{stall} = 47.9 \text{ N m s/rad}}$$

Question 4

Derive the transfer function that relates the input voltage to the angular velocity of the mixing element. Identify all parameter values in the transfer function.

We want the transfer function

$$\frac{\omega_{out}(s)}{V_{in}(s)}$$

for the selected 18 V EC-i 52 motor. The assignment states that:

- the shaft torsional stiffness is neglected
- only the motor inertia and mixing element inertia are considered
- gear and shaft inertias are ignored
- the load torque is viscous, $\tau = b\omega$
- the mixing element is a steel disk of diameter 3 cm and thickness 1 mm.

From the motor datasheet for the 18 V version:

- $R_m = 0.0555\ \Omega$
- $L_m = 0.0643$ mH $= 6.43 \times 10^{-5}$ H
- $K_T = 35.3$ mNm/A $= 0.0353$ N m/A
- $K_\omega = 270$ rpm/V
- $J_m = 170$ g cm^2
- nominal speed $= 4360$ rpm.

From Question 2,

$$N_g = \frac{\omega_{out}}{\omega_m}$$

and for the selected motor,

$$N_g = \frac{100}{4360} = 0.022936$$

So,

$$\omega_{out} = N_g \omega_m \Rightarrow \omega_m = \frac{\omega_{out}}{N_g}$$

Electrical equation,

$$V_{in} = R_m i + L_m \frac{di}{dt} + e_m$$

and

$$\omega_m = K_\omega e_m \Rightarrow e_m = \frac{\omega_m}{K_\omega}$$

Therefore,

$$V_{in} = R_m i + L_m \frac{di}{dt} + \frac{\omega_m}{K_\omega}$$

Since

$$\omega_m = \frac{\omega_{out}}{N_g}$$

we get

$$V_{in} = R_m i + L_m \frac{di}{dt} + \frac{\omega_{out}}{K_\omega N_g}$$

This is the electrical equation in terms of the output angular velocity.

Mechanical equation

The motor torque is

$$\tau_m = K_T i$$

Using the notes,

$$N_g = \frac{\omega_{out}}{\omega_m} = \frac{\tau_m}{\tau_{out}}$$

so

$$\tau_{out} = \frac{\tau_m}{N_g} = \frac{K_T}{N_g} i$$

At the output shaft, the load torque must balance inertia and damping:

$$\tau_{out} = J_{eq} \frac{d\omega_{out}}{dt} + b\omega_{out}$$

So the mechanical equation becomes

$$\frac{K_T}{N_g} i = J_{eq} \frac{d\omega_{out}}{dt} + b\omega_{out}$$

Equivalent inertia

The equivalent output-side inertia is

$$J_{eq} = J_L + \frac{J_m}{N_g^2}$$

where:

- J_L= inertia of the mixing element
- J_m= motor rotor inertia reflected to the output side

This reflection is correct because

$$\omega_{out} = N_g \omega_m$$

so the motor inertia appears larger at the slow output shaft by the factor $1/N_g^2$.

Compute the inertia values

- Motor inertia

The datasheet gives

$$J_m = 170 \ \text{g cm}^2$$

Convert to SI units:

$$1 \ \text{g cm}^2 = 10^{-7} \ \text{kg m}^2$$

so

$$J_m = 170 \times 10^{-7} = 1.70 \times 10^{-5} \qquad \text{kg m}^2$$

- Mixing element inertia

The mixing element is a steel disk with:

- diameter $= 3$ cm
- radius $r = 1.5$ cm $= 0.015$ m
- thickness $t = 1$ mm $= 0.001$ m.

Using steel density

$$\rho = 7850 \ \text{kg/m}^3$$

the disk volume is

$$V = \pi r^2 t$$
$$V = \pi(0.015)^2(0.001) = 7.069 \times 10^{-7} \ \text{m}^3$$

Mass:

$$m = \rho V$$
$$m = 7850(7.069 \times 10^{-7}) = 0.00555 \ \text{kg}$$

For a solid disk, the rotational inertia is

$$J_L = \frac{1}{2} m r^2$$
$$J_L = \frac{1}{2}(0.00555)(0.015)^2$$
$$J_L = 6.24 \times 10^{-7} \ \text{kg m}^2$$

Equivalent inertia

Using

$$N_g = 0.022936$$
$$J_{eq} = J_L + \frac{J_m}{N_g^2}$$

First compute

$$N_g^2 = (0.022936)^2 = 0.00052606$$

Then

$$\frac{J_m}{N_g^2} = \frac{1.70 \times 10^{-5}}{0.00052606} = 0.03231 \ \text{kg m}^2$$

Thus,

$$J_{eq} = 6.24 \times 10^{-7} + 0.03231 \approx 0.03231 \ \text{kg m}^2$$

So the reflected motor inertia dominates the total inertia.

Convert the equations to Laplace form

Electrical equation:

$$V_{in}(s) = (R_m + sL_m)I(s) + \frac{1}{K_\omega N_g}\omega_{out}(s)$$

Mechanical equation:

$$\frac{K_T}{N_g}I(s) = (J_{eq}s + b)\omega_{out}(s)$$

Solve the mechanical equation for $I(s)$:

$$I(s) = \frac{N_g}{K_T}(J_{eq}s + b)\omega_{out}(s)$$

Substitute into the electrical equation:

$$V_{in}(s) = (R_m + sL_m)\frac{N_g}{K_T}(J_{eq}s + b)\omega_{out}(s) + \frac{1}{K_\omega N_g}\omega_{out}(s)$$

Factor out $\omega_{out}(s)$:

$$V_{in}(s) = \omega_{out}(s)\left[\frac{N_g}{K_T}(R_m + sL_m)(J_{eq}s + b) + \frac{1}{K_\omega N_g}\right]$$

Therefore,

$$\boxed{\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{1}{\frac{N_g}{K_T}(R_m + sL_m)(J_{eq}s + b) + \frac{1}{K_\omega N_g}}}$$

This is the general transfer function, with b still symbolic.

Expanded standard form

Expand

$$(R_m + sL_m)(J_{eq}s + b) = L_m J_{eq}s^2 + (R_m J_{eq} + L_m b)s + R_m b$$

So the transfer function becomes

$$\boxed{\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{1}{\left(\frac{N_g}{K_T}L_m J_{eq}\right)s^2 + \left(\frac{N_g}{K_T}(R_m J_{eq} + L_m b)\right)s + \left(\frac{N_g}{K_T}R_m b + \frac{1}{K_\omega N_g}\right)}}$$

Substitute the known parameter values, but keep b symbolic

First convert K_ω to SI units.

Given

$$K_\omega = 270 \text{ rpm/V}$$
$$K_\omega = 270\left(\frac{2\pi}{60}\right) = 28.274 \text{ rad/s/V}$$

Now use:

$$R_m = 0.0555$$
$$L_m = 6.43 \times 10^{-5}$$
$$K_T = 0.0353$$
$$K_\omega = 28.274$$
$$N_g = 0.022936$$
$$J_{eq} = 0.03231$$

Now compute the recurring constants:

$$\frac{N_g}{K_T} = \frac{0.022936}{0.0353} = 0.64975$$
$$\frac{1}{K_\omega N_g} = \frac{1}{(28.274)(0.022936)} = 1.5407$$

Coefficient of s^2

$$a_2 = \frac{N_g}{K_T} L_m J_{eq}$$
$$a_2 = (0.64975)(6.43 \times 10^{-5})(0.03231)$$
$$a_2 = 1.35 \times 10^{-6}$$

Coefficient of s

$$a_1 = \frac{N_g}{K_T}(R_m J_{eq} + L_m b)$$
$$a_1 = 0.64975[(0.0555)(0.03231) + (6.43 \times 10^{-5})b]$$
$$a_1 = 0.64975[0.001793 + 6.43 \times 10^{-5} b]$$
$$a_1 = 0.001165 + 4.18 \times 10^{-5} b$$

Constant term

$$a_0 = \frac{N_g}{K_T} R_m b + \frac{1}{K_\omega N_g}$$
$$a_0 = (0.64975)(0.0555)b + 1.5407$$
$$a_0 = 0.03606b + 1.5407$$

So the transfer function is

$$\boxed{\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{1}{(1.35 \times 10^{-6})s^2 + (0.001165 + 4.18 \times 10^{-5} b)s + (1.5407 + 0.03606b)}}$$

For $b = 1.63$

Substitute

$$b = 1.63$$

into the symbolic transfer function.

Coefficient of s

$$0.001165 + 4.18 \times 10^{-5}(1.63) = 0.001233$$

Constant term

$$1.5407 + 0.03606(1.63) = 1.5995$$

So for $b = 1.63$,

$$\boxed{\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{1}{1.35 \times 10^{-6} s^2 + 0.001233s + 1.5995}}$$

Final answer

Using the motor data $R_m = 0.0555$, $L_m = 6.43 \times 10^{-5}$ H, $K_T = 0.0353$ N m/A, and $K_\omega = 28.274$ rad/s/V, the transfer function relating input voltage to output angular velocity is

$$\boxed{\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{1}{(1.35 \times 10^{-6})s^2 + (0.001165 + 4.18 \times 10^{-5} b)s + (1.5407 + 0.03606b)}}$$

If the damping coefficient is taken as $b = 1.63$, then the transfer function becomes

$$\boxed{\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{1}{1.35 \times 10^{-6} s^2 + 0.001233s + 1.5995}}$$

Normalized form

$$\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{7.41 \times 10^5}{s^2 + (863.0 + 30.96\, b)s + (1.141 \times 10^6 + 2.671 \times 10^4\, b)}$$

If b = 1.65

$$\boxed{\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{7.41 \times 10^5}{s^2 + 913.46\, s + 1.1845 \times 10^6}}$$

Question 5

MATHLAB was used

TransferFuctionCode1.m

```
b1 = 1.63;
b2 = 0.9*47.88;

num = 7.41e5;

den1 = [1, 863.0 + 30.96*b1, 1.141e6 + 2.671e4*b1];
den2 = [1, 863.0 + 30.96*b2, 1.141e6 + 2.671e4*b2];

G1 = tf(num, den1);
G2 = tf(num, den2);

poles1 = pole(G1);
zeros1 = zero(G1);
gain1 = dcgain(G1);

poles2 = pole(G2);
zeros2 = zero(G2);
gain2 = dcgain(G2);
```

Base system b= 1.65

$$G(s) = \frac{7.41 \times 10^5}{s^2 + 913.46\, s + 1.1845 \times 10^6}$$

Poles

The poles are the roots of

$$s^2 + 913.46\, s + 1.1845 \times 10^6 = 0$$

which gives

$$\boxed{s_{1,2} = -456.7 \pm 987.9\, j}$$

Zeros

The numerator is constant, so there are no finite zeros:

$$\boxed{\text{No zeros}}$$

System type

System type is determined by the number of poles at the origin. Since there is no factor of $s$in the denominator,

$$\boxed{\text{Type 0 system}}$$

Open-loop gain

The open-loop DC gain is

$$G(0) = \frac{7.41 \times 10^5}{1.1845 \times 10^6}$$

$$\boxed{G(0) \approx 0.626}$$

Increased damping, $b = (0.9 * \ b_{stall}\ )$

Now substitute

$$bstall = 43.092$$

into the general transfer function:

$$G(s) = \frac{7.41 \times 10^5}{s^2 + (863.0 + 30.96(43.092))s + (1.141 \times 10^6 + 2.671 \times 10^4(43.092))}$$

$$G(s) = \frac{7.41 \times 10^5}{s^2 + 2197.13\, s + 2.2920 \times 10^6}$$

Poles

The poles are the roots of

$$s^2 + 2197.13\, s + 2.2920 \times 10^6 = 0$$

which gives

$$\boxed{s_{1,2} = -1098.6 \pm 1041.7\, j}$$

Zeros

Again, the numerator is constant, so:

$$\boxed{\text{No zeros}}$$

System type

There are still no poles at the origin, so:

$$\boxed{\text{Type 0 system}}$$

Open-loop gain

$$G(0) = \frac{7.41 \times 10^5}{2.2920 \times 10^6}$$

$$\boxed{G(0) \approx 0.323}$$

Results

In both cases, the system has no zeros and is a Type 0 system. Increasing the damping moves the poles farther to the left in the complex plane, which means the response becomes more heavily damped and less oscillatory. At the same time, the open-loop gain decreases from about 0.626 to 0.323, so the steady-state output speed for a given input voltage is lower. This means that as the fluid resistance increases, the mixer becomes less responsive and produces a smaller final speed, although the response becomes less oscillatory and better damped. Since both pole pairs remain in the left half-plane, both cases are stable.

Question 6

Math Lab was used

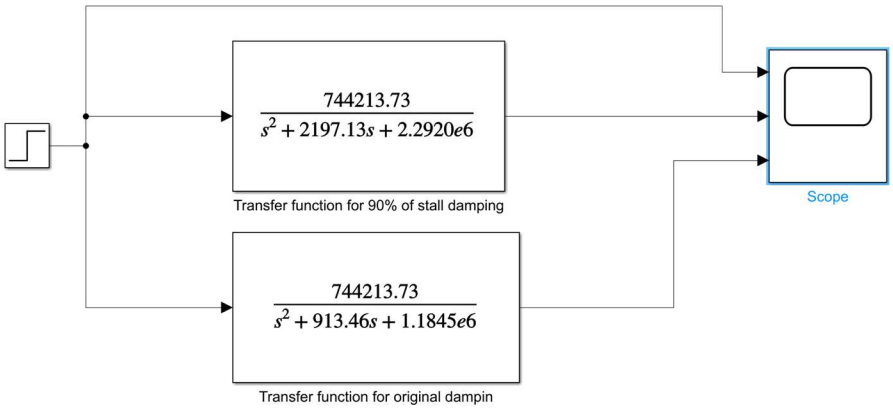

Figure 1 : Simulink format

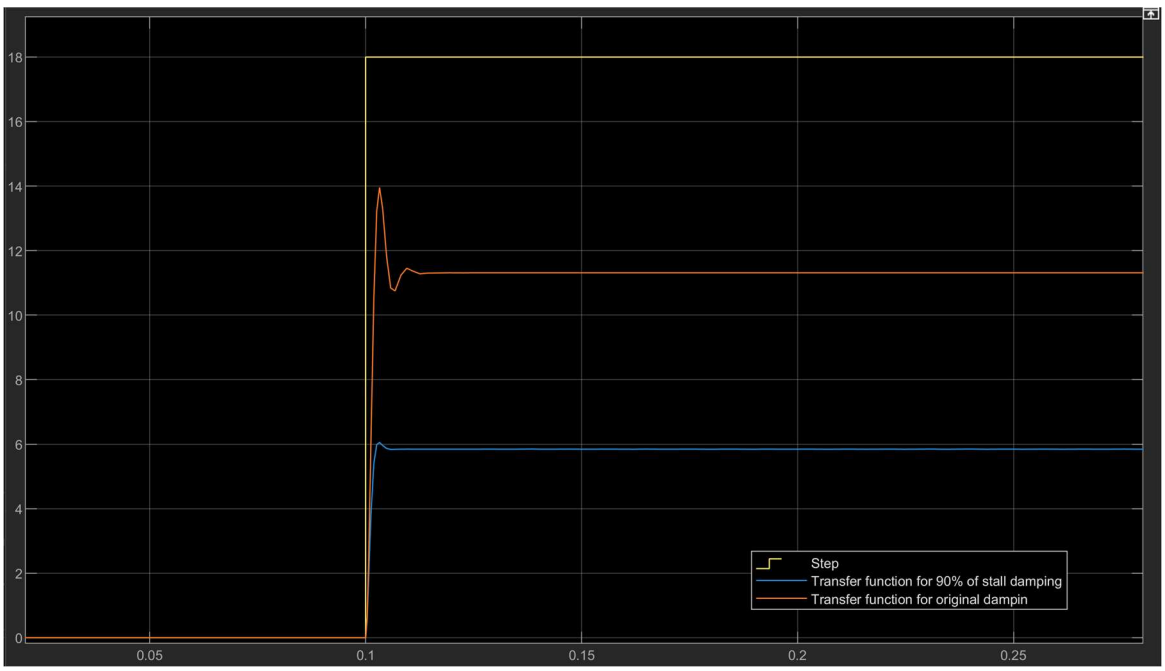

Figure 2 : Scope results for Step function

The step response was simulated in Simulink for a voltage input increasing from 0to the nominal motor voltage of 18 V. For the original damping case,

$$\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{744213.73}{s^2 + 913.46s + 1.1845 \times 10^6}$$

while for the $0.9b_{stall}$case,

$$\frac{\omega_{out}(s)}{V_{in}(s)} = \frac{744213.73}{s^2 + 2197.13s + 2.2920 \times 10^6}$$

Using the Final Value Theorem, the steady-state output speeds are approximately 11.31 rad/s for the original damping and 5.85 rad/s for the $0.9b_{stall}$case. The Simulink results confirm this. The original damping response has a larger overshoot and a more oscillatory transient, with a peak near 14 rad/s, while the $0.9b_{stall}$ response is smoother and has very little overshoot. Both systems settle rapidly, but the increased damping substantially reduces the final mixer speed. This demonstrates the trade-off between transient smoothness and steady-state performance: higher damping improves stability and reduces overshoot but lowers the final operating speed.

Calculation

We use the two transfer functions already obtained for Question 4:

For the original damping $b = 1.63$,

$$G_1(s) = \frac{\omega_{out}(s)}{V_{in}(s)} = \frac{744213.73}{s^2 + 913.46s + 1.1845 \times 10^6}$$

For the 90% stall damping case, $b = 43.092$,

$$G_2(s) = \frac{\omega_{out}(s)}{V_{in}(s)} = \frac{744213.73}{s^2 + 2197.13s + 2.2920 \times 10^6}$$

For a step input from 0 to 18 V,

$$V_{in}(s) = \frac{18}{s}$$

So the outputs are:

$$\Omega_1(s) = G_1(s)\frac{18}{s} \text{and} \Omega_2(s) = G_2(s)\frac{18}{s}$$

Final value

Using the Final Value Theorem,

$$\omega_{ss} = \lim_{s \to 0} s\Omega(s) = 18\, G(0)$$

Original damping

$$G_1(0) = \frac{744213.73}{1.1845 \times 10^6} = 0.6283$$

so

$$\omega_{ss,1} = 18(0.6283) = 11.31 \text{ rad/s}$$

90% of stall damping

$$G_2(0) = \frac{744213.73}{2.2920 \times 10^6} = 0.3247$$

so

$$\omega_{ss,2} = 18(0.3247) = 5.845 \text{ rad/s}$$

So the original damping case settles at a higher speed, while the larger damping case settles at a lower speed, which is physically what we expect.

Standard second-order form

Compare each denominator to

$$s^2 + 2\zeta\omega_n s + \omega_n^2$$

Original damping

$$\omega_{n,1} = \sqrt{1.1845 \times 10^6} = 1088.35 \text{ rad/s}$$
$$2\zeta_1\omega_{n,1} = 913.46$$
$$\zeta_1 = \frac{913.46}{2(1088.35)} = 0.4197$$

90% of stall damping

$$\omega_{n,2} = \sqrt{2.2920 \times 10^6} = 1513.94 \text{ rad/s}$$
$$2\zeta_2\omega_{n,2} = 2197.13$$
$$\zeta_2 = \frac{2197.13}{2(1513.94)} = 0.7256$$

Since both damping ratios are less than 1, both responses are underdamped, but the second case is much more heavily damped.

3. Percent overshoot

For a standard underdamped second-order system,

$$\%OS = 100e^{-\pi\ /\sqrt{1-\zeta^2}}$$

Original damping

With $\zeta_1 = 0.4197$,

$$\%OS_1 \approx 23.4\%$$

90% of stall damping

With $\zeta_2 = 0.7256$,

$$\%OS_2 \approx 3.6\%$$

So the original damping case has a much more noticeable overshoot, while the 90% stall damping case has only a small overshoot.

Question 7

MATLAB was used

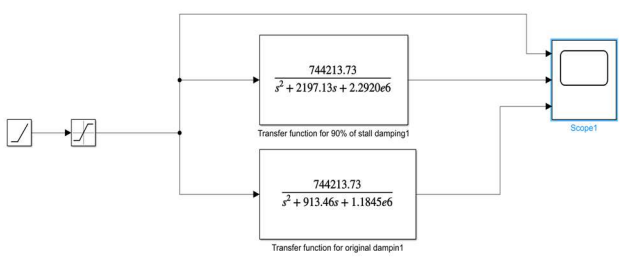

Figure 3 : Simulink format

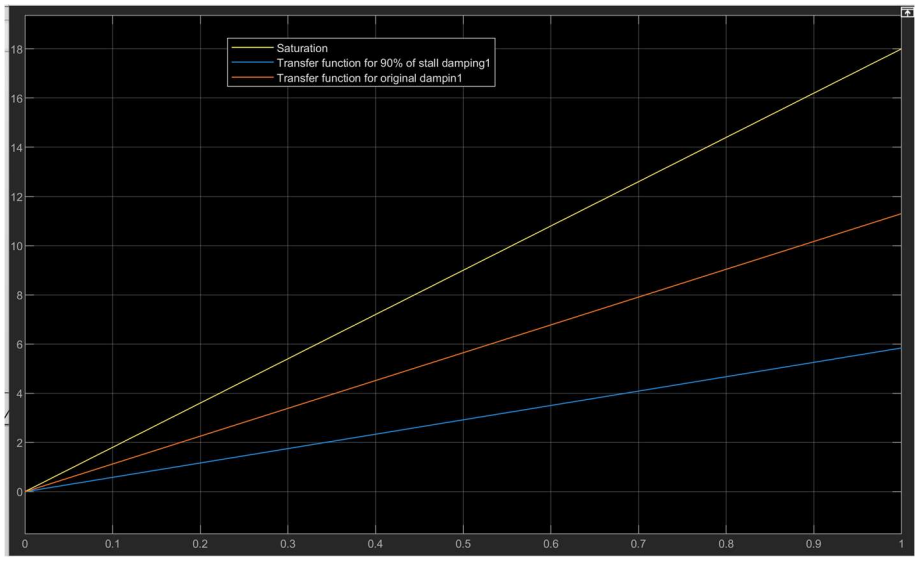

Figure 4 : Scope results for Ramp function

The system response was simulated for a ramp input in which the voltage increases linearly from 0 V to the nominal voltage of 18 V over 1 second. Since the selected motor is the 18 V version, this matches the assignment requirement.

The Simulink results show that both responses rise smoothly and almost linearly with the input voltage. Unlike the step-response case, there is essentially no noticeable overshoot or oscillation in the plotted ramp response. This occurs because the input increases gradually over 1 second, while the motor system dynamics are much faster than that, so the output can track the changing input closely.

For the original damping case, the output increases more rapidly and reaches about 11.3 rad/s by the time the input reaches 18 V at $t = 1$ s. For the 90% of stall damping case, the response is lower throughout and reaches about 5.85 rad/s at $t = 1$ s. Thus, the original damping case again produces higher mixer speed, while the increased damping substantially reduces the final speed.

Compared with the step-input simulation, the ramp-input response is much smoother for both cases. In the step-response case, the original damping system showed a noticeable overshoot and a short transient oscillation, while the $0.9b_{stall}$case was more heavily damped. In contrast, under ramp input both responses increase gradually and behave in a much more controlled manner. The same overall trend remains: the original damping case gives better speed performance, whereas the 90% of stall damping case gives a lower but more restrained response due to the heavier effective load.

Question 8

If the motor terminals are placed in an open circuit, current cannot flow, so the motor produces little electromagnetic braking torque and simply coasts to a stop under mechanical and fluid damping. This results in a relatively slow stop. If the terminals are short-circuited, the back EMF generated by the spinning motor drives current through the windings, producing a torque opposite to the rotation. This creates dynamic breaking and stops the motor much faster. Therefore, a short circuit is more effective for rapid stopping, but it may also produce high braking currents and additional heating. An open circuit is safer electrically, but much less effective for fast deceleration.

Question 9

The selected EC-i 52 brushless DC motor is capable of driving the mixer, but its suitability depends strongly on the viscous damping of the load. Under the original damping condition, the simulated steady-state mixer speed was approximately 11.31 rad/s, which is about 108 rpm, so the motor is able to meet the intended operating range. However, when the damping was increased to $0.9b_{stall}$, the steady-state speed dropped to about 5.85 rad/s, or 56 rpm, showing a substantial reduction in performance. This indicates that the motor is acceptable for moderate conditions but may be marginal for highly viscous fluids or heavier loading. Improvements could include selecting a motor with a higher continuous torque, choosing a gearbox with greater torque multiplication, or implementing closed-loop speed control to maintain the desired mixer speed despite changes in fluid resistance. Overall, the motor is adequate for the task, but it is not the most robust choice if large variations in viscous load are expected.